Lemma: 对$\forall n\in\mathbb{Z}_{\geqslant 1}$，若不存在$k\in\mathbb{Z}$，使得$k^2=n$，则有：$\sqrt{n}\in\mathbb{R}\setminus\mathbb{Q}$

Proof: $\because n\in\mathbb{Z}_{\geqslant 1}$ $\therefore n\in\mathbb{Z}$且$n>0$ $\therefore \sqrt{n}\in\mathbb{R}$

假设$\sqrt{n}\in\mathbb{Q}$，则有：$\exists p,q\in\mathbb{Z}$，$q\neq 0$，$\gcd(p,q)=1$，使得$\sqrt{n}=\frac{p}{q}$

$\therefore p=q\sqrt{n}$ $\therefore p^2=q^2n$

$\because p^2\in\mathbb{Z}$，$q^2\in\mathbb{Z}$且$q^2\neq 0$，$n\in\mathbb{Z}$ $\therefore q^2\mid p^2$

$\because q^2\in\mathbb{Z}$，$p^2\in\mathbb{Z}$，$q^2\neq 0$，$q^2\mid p^2$ $\therefore \gcd(q^2,p^2)=|q^2|=q^2$

$\therefore \gcd(p^2,q^2)=\gcd(q^2,p^2)=q^2$

$\because p,q\in\mathbb{Z}$，$2\in\mathbb{Z}_{\geqslant 1}$ $\therefore \gcd(p^2,q^2)=(\gcd(p,q))^2=1^2=1$

$\therefore q^2=1$ $\therefore p^2=q^2n=1\cdot n=n$

$\therefore p\in\mathbb{Z}$且$p^2=n$．矛盾． $\therefore \sqrt{n}\notin\mathbb{Q}$ $\therefore \sqrt{n}\in\mathbb{R}\setminus\mathbb{Q}$ $\square$

Lemma: $D\in\mathbb{Z}$是无平方因子的非零正整数，$D\neq 1$，则有：$\sqrt{D}\in\mathbb{R}\setminus\mathbb{Q}$

Proof: $\because D\in\mathbb{Z}$是正整数 $\therefore D\in\mathbb{Z}_{\geqslant 1}$

假设 存在$k\in\mathbb{Z}$，使得$k^2=D$．则有：$\because D\neq 0$ $\therefore k\neq 0$ $\therefore k^2\neq 0$

$\because D=k^2\cdot 1$，$D\in\mathbb{Z}$，$k^2\in\mathbb{Z}$，$k^2\neq 0$，$1\in\mathbb{Z}$ $\therefore k^2\mid D$

$\therefore D$有平方因子．矛盾． $\therefore$不存在$k\in\mathbb{Z}$，使得$k^2=D$

$\therefore \sqrt{D}\in\mathbb{R}\setminus\mathbb{Q}$ $\square$

3.1 练习

设 $D\in\mathbb{Z}$ 是无平方因子的非零整数，$D\neq 1$.

(i) 验证 $\mathbb{Q}(\sqrt{D}):=\{a+b\sqrt{D}\in\mathbb{C}: a,b\in\mathbb{Q}\}$ 是复数域 $\mathbb{C}$ 的子域，称为二次域. 具体说明非零元的求逆公式.

Proof: 对 $\forall D\neq 0$，有:

若 $D>0$，则有 $\sqrt{D}\in\mathbb{R}\subseteq\mathbb{C}$

若 $D<0$，则有: $\sqrt{D}=\sqrt{-(-D)}=\sqrt{(-1)(-D)}=\sqrt{-1}\cdot\sqrt{-D}=\sqrt{-D}\cdot i\in\mathbb{C}$

$\therefore \sqrt{D}\in\mathbb{C}$

任取 $\mathbb{Q}(\sqrt{D})$ 中的一个元素: $a+b\sqrt{D}$（其中 $a,b\in\mathbb{Q}$）.

$\because a\in\mathbb{Q}\subseteq\mathbb{C}$，$b\in\mathbb{Q}\subseteq\mathbb{C}$，$\sqrt{D}\in\mathbb{C}$ $\quad\therefore a+b\sqrt{D}\in\mathbb{C}$

$\therefore \mathbb{Q}(\sqrt{D})\subseteq\mathbb{C}$

$\because 0=0+0\sqrt{D}\in\mathbb{Q}(\sqrt{D})$ $\quad\therefore \mathbb{Q}(\sqrt{D})$ 是 $\mathbb{C}$ 的非空子集.

$\because 0=0+0\sqrt{D}\in\mathbb{Q}(\sqrt{D})$，$1=1+0\sqrt{D}\in\mathbb{Q}(\sqrt{D})$

对 $\forall\, a+b\sqrt{D},\ c+d\sqrt{D}\in\mathbb{Q}(\sqrt{D})$（其中 $a,b,c,d\in\mathbb{Q}$），有:

$(a+b\sqrt{D})-(c+d\sqrt{D})=a+b\sqrt{D}-c-d\sqrt{D}=(a-c)+(b-d)\sqrt{D}\in\mathbb{Q}(\sqrt{D})$

对 $\forall\, a+b\sqrt{D}\in\mathbb{Q}(\sqrt{D})$，$c+d\sqrt{D}\in\mathbb{Q}(\sqrt{D})\backslash\{0\}$（其中 $a,b,c,d\in\mathbb{Q}$），有:

假设 $c=0$ 且 $d=0$，则有: $c+d\sqrt{D}=0+0\sqrt{D}=0$ 矛盾.

$\therefore c\neq 0$ 或 $d\neq 0$.

$\because c\in\mathbb{Q}$，$d\in\mathbb{Q}$，$D\in\mathbb{Z}\subseteq\mathbb{Q}$ $\quad\therefore c^2-d^2D\in\mathbb{Q}$.

假设 $c^2-d^2D=0$，则有:

若 $d=0$，则有: $0=c^2-d^2D=c^2-0\cdot D=c^2-0=c^2$ $\quad\therefore c=0$

$\therefore c=0$ 且 $d=0$. 矛盾. $\quad\therefore d\neq 0$ $\quad\therefore c^2=d^2D$

$\because d\neq 0 \quad \therefore d^2\neq 0 \quad \therefore D=\frac{c^2}{d^2}=\left(\frac{c}{d}\right)^2\geqslant 0 \quad \because D\neq 0 \quad \therefore D>0$

$\therefore D\in\mathbb{Z}_{\geqslant 1} \quad \therefore D\in\mathbb{Z}$是无平方因子的非零正整数，$D\neq 1 \quad \therefore \sqrt{D}\in\mathbb{R}\setminus\mathbb{Q}$

$\because \sqrt{D}=\sqrt{\left(\frac{c}{d}\right)^2}=\left|\frac{c}{d}\right|\in\mathbb{Q}$．矛盾 $\quad \therefore c^2-d^2D\neq 0$

$\therefore c^2-d^2D\in\mathbb{Q}$ 且 $c^2-d^2D\neq 0$

$\therefore \frac{c}{c^2-d^2D}\in\mathbb{Q},\ \frac{-d}{c^2-d^2D}\in\mathbb{Q} \quad \therefore \frac{c}{c^2-d^2D}+\frac{-d}{c^2-d^2D}\sqrt{D}\in\mathbb{Q}(\sqrt{D})$

$\because (c+d\sqrt{D})\left(\frac{c}{c^2-d^2D}+\frac{-d}{c^2-d^2D}\sqrt{D}\right)$

$=\frac{c^2}{c^2-d^2D}+\frac{-cd}{c^2-d^2D}\sqrt{D}+\frac{cd}{c^2-d^2D}\sqrt{D}+\frac{-d^2D}{c^2-d^2D}$

$=\frac{c^2-d^2D}{c^2-d^2D}=1$

$\because c+d\sqrt{D}\in\mathbb{Q}(\sqrt{D})\subseteq\mathbb{C},\ \frac{c}{c^2-d^2D}+\frac{-d}{c^2-d^2D}\sqrt{D}\in\mathbb{Q}(\sqrt{D})\subseteq\mathbb{C}$

$\therefore (c+d\sqrt{D})^{-1}=\frac{c}{c^2-d^2D}+\frac{-d}{c^2-d^2D}\sqrt{D}\in\mathbb{Q}(\sqrt{D})$

$\therefore (a+b\sqrt{D})\cdot(c+d\sqrt{D})^{-1}=(a+b\sqrt{D})\left(\frac{c}{c^2-d^2D}+\frac{-d}{c^2-d^2D}\sqrt{D}\right)$

$=\frac{ac-bdD}{c^2-d^2D}+\frac{-ad+bc}{c^2-d^2D}\sqrt{D}\in\mathbb{Q}(\sqrt{D})$

$\therefore \mathbb{Q}(\sqrt{D})$是$\mathbb{C}$的子域．

对$\forall\ a+b\sqrt{D}\in\mathbb{Q}(\sqrt{D})$，$a+b\sqrt{D}\neq 0$，有：$(a,b\in\mathbb{Q})$

假设 $a=0$且$b=0$．则有：$a+b\sqrt{D}=0+0\sqrt{D}=0$，矛盾．$\therefore a\neq 0$ 或 $b\neq 0$

假设 $a^2-b^2D=0$，则有：若$b=0$，则 $0=a^2-b^2D=a^2-0D=a^2 \quad \therefore a=0$

$\therefore a=0$且$b=0$ 矛盾．$\therefore b\neq 0 \quad \because b\in\mathbb{Q} \quad \therefore b^2\neq 0 \quad \therefore a^2=b^2D \quad \therefore D=\frac{a^2}{b^2}$

$\therefore D = \frac{a^2}{b^2} = \left(\frac{a}{b}\right)^2 \quad \because a \in \mathbb{Q}, b \in \mathbb{Q}, \quad b \neq 0 \quad \therefore \frac{a}{b} \in \mathbb{Q}$

$\therefore D = \left(\frac{a}{b}\right)^2 \geqslant 0 \quad \because D \neq 0 \quad \therefore D > 0$ $\therefore D$是无平方因子的非零正整数，$D \neq 1$

$\therefore \sqrt{D} \in \mathbb{R} \setminus \mathbb{Q}$. $\because \sqrt{D} = \sqrt{\left(\frac{a}{b}\right)^2} = \left|\frac{a}{b}\right| \in \mathbb{Q}$ $\therefore$矛盾.

$\therefore a^2 - b^2 D \neq 0 \quad \therefore a^2 - b^2 D \in \mathbb{Q}$ 且 $a^2 - b^2 D \neq 0$

$\therefore \frac{a}{a^2 - b^2 D} + \frac{-b}{a^2 - b^2 D}\sqrt{D} \in \mathbb{Q}(\sqrt{D})$，且有:

$$(a + b\sqrt{D})\left(\frac{a}{a^2 - b^2 D} + \frac{-b}{a^2 - b^2 D}\sqrt{D}\right) = 1$$

$$\therefore (a + b\sqrt{D})^{-1} = \frac{a}{a^2 - b^2 D} + \frac{-b}{a^2 - b^2 D}\sqrt{D} \in \mathbb{Q}(\sqrt{D})$$

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